

Name
Class
R. No.

• Mathematics
• IIT JEE
• Test 7(August 12, 2010)
• Inverse Trigonometry (NCERT)
• Maximum Marks 110
• Time 60 minutes

Instructions

1. The test contains 7 questions.
2. The marking scheme will include marks for partially correct answers also.

1. Prove the following:

$$3\sin^{-1} x = \sin^{-1}(3x - 4x^3), x \in \left[\frac{1}{2}, 1\right]$$

10 Marks

2. Prove the following:

$$2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$$

10 Marks

3. Write the following functions in the simplest form:

i) $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}, x \neq 0$

10 Marks

ii) $\tan^{-1} \frac{x}{\sqrt{a^2-x^2}}, |x| < a$

10 Marks

4. Find the values of the following:

$$\tan \frac{1}{2} \left(\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right), |x| < 1, y > 0, xy < 1$$

20 Marks

5. If $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$, then find the value of x.

20 Marks

6. Show that $\sin^{-1} \frac{3}{5} - \sin^{-1} \frac{8}{17} = \cos^{-1} \frac{84}{85}$.

15 Marks

7. Solve $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$

15 Marks